

Exercise 3.6 (*)

Suppose $f : (a, b) \rightarrow \mathbb{R}$ is uniformly continuous, and $(x_j)_{j=0}^\infty, (y_j)_{j=0}^\infty \subset (a, b)$ satisfy $x_j \rightarrow a$ and $y_j \rightarrow b$.

- (a) Show that the sequences $(f(x_j))_{j=0}^\infty$ and $(f(y_j))_{j=0}^\infty$ are Cauchy.

Solution. Given $\varepsilon > 0$ choose $\delta > 0$ such that $|u - v| < \delta \Rightarrow |f(u) - f(v)| < \varepsilon$ (uniform continuity). Since $x_j \rightarrow a$ and $y_j \rightarrow b$, both (x_j) and (y_j) are Cauchy. Hence there is N with $|x_j - x_k| < \delta$ and $|y_j - y_k| < \delta$ for all $j, k \geq N$, which implies $|f(x_j) - f(x_k)| < \varepsilon$ and $|f(y_j) - f(y_k)| < \varepsilon$ for $j, k \geq N$. Thus both image sequences are Cauchy.

- (b) Setting $f_L := \lim_{j \rightarrow \infty} f(x_j)$ and $f_R := \lim_{j \rightarrow \infty} f(y_j)$, show that

$$\tilde{f} : [a, b] \rightarrow \mathbb{R}, \quad \tilde{f}(x) = \begin{cases} f_L, & x = a, \\ f(x), & a < x < b, \\ f_R, & x = b, \end{cases}$$

is continuous on $[a, b]$.

Solution. By (a) the limits f_L, f_R exist (Cauchy in $\mathbb{R}$ hence convergent). On (a, b) , $\tilde{f} = f$, so continuity holds. We check continuity at a ; the case b is analogous.

Let $\varepsilon > 0$. Pick $\delta > 0$ for uniform continuity of f and choose J large so that $|x_J - a| < \delta/2$ and $|f(x_J) - f_L| < \varepsilon$. If $0 < |t - a| < \delta/2$, then $|t - x_J| \leq |t - a| + |x_J - a| < \delta$, hence

$$|\tilde{f}(t) - \tilde{f}(a)| = |f(t) - f_L| \leq |f(t) - f(x_J)| + |f(x_J) - f_L| < \varepsilon + \varepsilon = 2\varepsilon.$$

Therefore $\tilde{f}$ is continuous at a . A symmetric argument using (y_j) gives continuity at b . Thus $\tilde{f}$ is continuous on $[a, b]$.

- (c) Deduce that f is uniformly continuous on (a, b) if and only if there exists a continuous $\tilde{f} : [a, b] \rightarrow \mathbb{R}$ with $\tilde{f}(x) = f(x)$ for all $x \in (a, b)$ (i.e. $\tilde{f}$ is an extension of f).

Solution. ($\Rightarrow$) If f is uniformly continuous, parts (a)–(b) construct a continuous extension $\tilde{f}$ on $[a, b]$.

($\Leftarrow$) If such $\tilde{f}$ exists, then $\tilde{f}$ is continuous on the compact set $[a, b]$, hence uniformly continuous. Its restriction to (a, b) equals f and is therefore uniformly continuous.

Exercise 3.5 (*)

Let $E \subset \mathbb{R}^n$ be compact. Define $C^0(E, \mathbb{R}^m)$ to be the set of all continuous functions $f : E \rightarrow \mathbb{R}^m$, equipped with the norm

$$\|f\|_{C^0} := \sup_{x \in E} \|f(x)\| \in [0, \infty).$$

Show that $C^0(E, \mathbb{R}^m)$ is complete.

Solution. Let (f_k) be a Cauchy sequence in the sup norm $\|\cdot\|_{C^0}$. Then, for every $x \in E$,

$$\|f_k(x) - f_\ell(x)\| \leq \|f_k - f_\ell\|_{C^0} \xrightarrow{k, \ell \rightarrow \infty} 0,$$

so for each x the sequence $(f_k(x))$ is Cauchy in $\mathbb{R}^m$ and hence converges. Define $f(x) := \lim_{k \rightarrow \infty} f_k(x)$; this gives a function $f : E \rightarrow \mathbb{R}^m$.

Uniform convergence. Given $\varepsilon > 0$, choose N so that $\|f_k - f_\ell\|_{C^0} < \varepsilon$ for all $k, \ell \geq N$. Fix $\ell \geq N$ and let $k \rightarrow \infty$; by the pointwise limit we obtain

$$\|f - f_\ell\|_{C^0} = \sup_{x \in E} \|f(x) - f_\ell(x)\| \leq \varepsilon.$$

Thus $f_k \rightarrow f$ uniformly on E . In particular, $\|f\|_{C^0} \leq \|f - f_\ell\|_{C^0} + \|f_\ell\|_{C^0} < \infty$, so f is bounded.

Continuity of the limit. Let $x \in E$ and $\varepsilon > 0$. Choose $\ell \geq N$ so that $\|f - f_\ell\|_{C^0} < \varepsilon/3$. Since f_ℓ is continuous at x , there exists $\delta > 0$ with $\|y - x\| < \delta \Rightarrow \|f_\ell(y) - f_\ell(x)\| < \varepsilon/3$. Then for $\|y - x\| < \delta$,

$$\|f(y) - f(x)\| \leq \|f(y) - f_\ell(y)\| + \|f_\ell(y) - f_\ell(x)\| + \|f_\ell(x) - f(x)\| < \varepsilon.$$

Hence f is continuous on E , i.e. $f \in C^0(E, \mathbb{R}^m)$.

Convergence in norm. With the same N as above, fix $\ell \geq N$. For $k \geq N$,

$$\|f_k - f\|_{C^0} \leq \|f_k - f_\ell\|_{C^0} + \|f_\ell - f\|_{C^0} \leq \varepsilon + \varepsilon = 2\varepsilon,$$

so letting $\varepsilon \downarrow 0$ (or starting with $\varepsilon/2$) gives $\|f_k - f\|_{C^0} \rightarrow 0$.

Therefore every Cauchy sequence in $C^0(E, \mathbb{R}^m)$ converges (in $\|\cdot\|_{C^0}$) to an element of $C^0(E, \mathbb{R}^m)$. Thus $C^0(E, \mathbb{R}^m)$ is complete.